



Inception of a Spaceflight-specific Mouse to Human Expression Profiling Translation Model

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Abstract

Rodents are foundational model organisms often utilized due to their seemingly analogous morphologies and biological responses to humans. However, recent studies have demonstrated that murine model data are limited in their applicability, particularly in inflammatory disease. In space studies, accurately predicting human response from mouse data is critical due to extreme limiting factors in both rodent and human spaceflight research. With successful prediction, spaceflight ailments can be predicted and prevented while respecting the constraints of the spaceflight industry and minimizing danger to humans. To do so, novel methodologies must be developed that predict human response from murine data after considering biological differences between rodents and humans in spaceflight.

After considering terrestrial models, we determined that a spaceflight-based expression profiling translation tool should be created to accurately capture predictions of human gene expression in spaceflight from mouse data. To prepare to build this model, we organized known human spaceflight risks, those analog human diseases as training data categories, then identified existing RNASeq disease datasets from GEO as potential training data. In addition, we classified existing GeneLab mouse differential gene expression datasets for use as experimental data.

Background and Introduction

Recent studies have demonstrated the limitations of murine model data. Scores of effective drugs developed in mice have failed in humans, and one study found murine inflammatory gene expression direction correlation to humans to be almost random [1]. To this end, researchers have developed expression profiling translation models to more effectively predict human gene expression response to health conditions from mouse data.

One such development is terrestrial-based translation model *Found in Translation (FIT)* [2]. *FIT* was created using transcription profiling microarray and RNASeq datasets that paired tissue-matched diseases between mice and humans as training data. Through usage of machine learning, the model predicts human effect size from input mouse gene expression data by considering differences in gene expression between species in the same disease.

We used *FIT*'s built-in classifier with a reported 80% success rate to predict if our mouse spaceflight data were likely to be improved by the model. In all cases, our data were stated to be unlikely to be improved. To probe this prediction, we ran GeneLab's RR1 datasets (GLDS-98 to GLDS-105) through *FIT*. We found that *FIT* returned unlikely results: upregulation of almost all genes, and even more so when only genes within the 30 percentile CI were included.

Given these results, we determined that *FIT* was not applicable to our data. More broadly, we realized that terrestrial-based expression profiling models do not capture crucial information about spaceflight stressors that could influence gene expression changes. We therefore determined that a new, spaceflight-specific model should be created and began to generate input data for it.

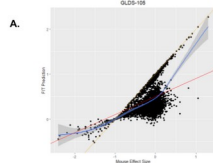


Figure A: Comparison of GLDS-105 (Tibialis Anterior) RNASeq effect sizes before and after *FIT*. Comparing our mouse effect size to *FIT*'s predicted human effect size, we found many genes to have an increased, but scaled, effect size. Approximately 90% of genes were predicted to be upregulated after being run through *FIT*.

Results

We determined the need for a new space-based expression profiling model and began its creation by gathering and organizing datasets relevant to expression profiling changes in spaceflight.

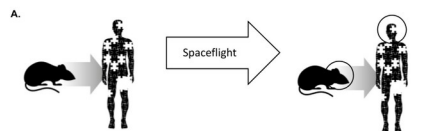


Figure A: The premise of the new expression profiling translation model. Building on existing models that predicted human effect based on mouse expression profiling data in inflammatory disease, we considered how to predict human response from mouse RNASeq data after spaceflight. Modified from [2].

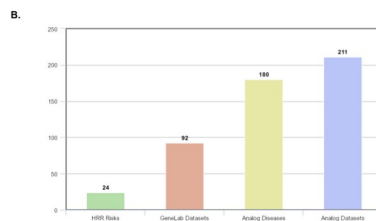


Figure B: A summary of the information collected for data input into the model. 24 HRP risks and 92 GeneLab datasets were used to predict 180 terrestrial diseases relevant to spaceflight. From these diseases, 211 RNASeq human datasets were chosen for use as input data.

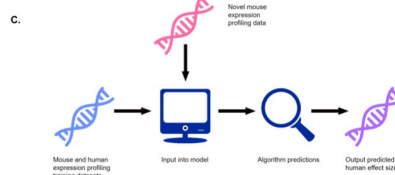


Figure C: An overview of the steps necessary in developing our expression profiling model. Tissue-matched expression profiling datasets paired between mice and humans relevant to spaceflight ailments are used as training data. Machine learning algorithms synthesize the differences between healthy and disease states between the two organisms, and when presented with novel mouse expression profiling data before and after spaceflight, predict human effect size.

Materials and Methods

To gather input data, we first identified known biological spaceflight risks from NASA's Human Research Roadmap (HRR) website. We characterized all published mouse and human GeneLab transcription profiling datasets for potential downstream use. Spaceflight studies were noted due to their innate utility. We then classified internal information within the HRR webpage and external data through Google search to choose approximately 180 terrestrial diseases that could be useful terrestrial models for these risks. We then utilized web platform GREIN to identify human RNASeq disease datasets matching our previously identified risks that compared disease state to control state. In total, we identified 211 datasets that could be useful as model training data. In addition, we paired GeneLab human and mouse translation profiling datasets to each GREIN dataset for later usage as training or experimental data.

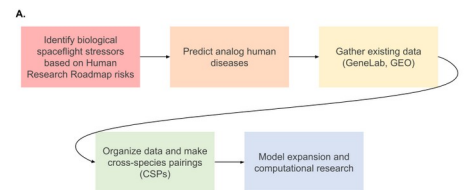


Figure A: A diagram of how datasets were gathered for the model and the direction of future work.

Future Directions

Future research will complete and expand data input and also further define the model. To complete data input, mouse RNASeq datasets should be classified as analogs to the identified human disease datasets and, in line with *FIT*, cross-species pairings should be tissue-matched. The model's data can be further expanded by including datasets studying all types of cancer, as well as microarray, single-cell and noncoding expression profiling datasets. As well, additional tools and databases such as ReGEO can be utilized to find additional studies.

When finalizing chosen model input data, investigators should consider the severity and importance of individual diseases. Specifically, diseases should be classified by their development time, as a model disease which takes years to develop may not be representative of short-duration spaceflight ailments. Furthermore, future disease selection should utilize published articles in addition to search engine results. This would allow for both proposed and identified disease analogs to be included in the model.

After finalizing the input data of the model, computational work must be conducted. Investigators must determine significant computational parameters, relevant machine learning algorithms, and how to validate the model, in addition to considering interface and publication design. Collaboration with *FIT* creators is encouraged as a method to build upon existing work.

[1] Seok, et al. Genomic responses in mouse models poorly mimic human inflammatory diseases. *Proceedings of the National Academy of Sciences*, 9, 3507–3512 (2013). <https://doi.org/10.1073/pnas.1222878110>

[2] Normand, R., Du, W., Briller, M. et al. Found In Translation: a machine learning model for mouse-to-human inference. *Nat Methods* 15, 1067–1073 (2018). <https://doi.org/10.1038/s41592-018-0214-93>

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